Rosenbrock · raw-gradient and momentum norms

Noisy raw gradient $\|g_t\|$ — Stored momentum $\|m_t\|$

(a) $\eta = 10^{-2}$ (b) $\eta = 3.1623 \times 10^{-3}$ (c) $\eta = 10^{-3}$

(d) (e) (f)

(g) (h) (i)

(j) (k) (l)

(m) (n) (o)

$\sigma = 0$ Vector norm

$\sigma = 0.01$ Vector norm

$\sigma = 0.1$ Vector norm

$\sigma = 1$ Vector norm

$\sigma = 10$ Vector norm

10^{2} 10^{1} 10^{0} 10^{-1} 10^{-2} 10^{-3}

250 500 750 1000 1250 1500 1750 2000

Optimizer update t

Every update, no averaging or thinning; shared log Y limits. Columns: learning rate; rows: gradient-coordinate noise standard deviation.

AdamW, start (0,1), decay 0.1; one paired noise seed1234. Sigma0 reuses deterministic controls; momentum is before bias correction/adaptive scaling.